AADIT AGRAWAL

Computer Science Engineer
AI/ML · HPC · DevOps · Security

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LANGUAGES
Python, C/C++, TypeScript, JavaScript, Rust, Go, Zig, Bash/Zsh, SQL, Java, Kotlin, C#, Lua, Swift, Haskell, Nix, LaTeX, Typst

FRONTEND
Next.js, React, Svelte, Vue, Astro, Tailwind, Radix, shadcn/ui

BACKEND
Node.js, Bun, Deno, Express, tRPC, FastAPI, Flask, Django

DEVOPS
Docker, Proxmox, Linux, Git, GitHub Actions, Ansible, Terraform, Pulumi

NETWORKING / MONITORING
Tailscale, WireGuard, Nginx, Caddy, Traefik, HAProxy, Grafana, Prometheus, Loki

CLOUD
AWS, GCP, Azure, Cloudflare Workers, Vercel, Fly.io, Supabase, Convex

AI / ML
PyTorch, JAX, Transformers, DeepSpeed, FSDP, Ray, W&B, OpenCV, YOLO, Detectron2, LangChain, Haystack

LLM / HPC
SGLang, vLLM, llama.cpp, TensorRT-LLM, MLX, Ollama, CUDA, RDMA, NCCL

AGENTIC TOOLING
Claude Code, Codex, Pi.dev, Droid, Cursor, opencode, ampcode, CodeRabbit

DATABASES
PostgreSQL, MariaDB, MongoDB, SQLite, Redis, ChromaDB, pgvector, Milvus

ARCHITECTURE
ARM64, Apple Silicon, Jetson, AMD64/x86-64, RISC-V, SIMD

SECURITY
Wireshark, Burp, Nmap, Metasploit, Ghidra

EDUCATION

Manipal Institute of Technology · *BTech — CCE, MAHE* — Jul 2023 – May 2027

EXPERIENCE

Head Student Researcher · *CASIR, MAHE* · *ecitis.org* — May 2025 – Present

- Architected **HPC cluster** (96-core Threadripper 7995WX, 48GB A6000, 256GB RAM) — **1,500+ daily inference req, 99.2% uptime.**
- Leading the setup of a cluster **$200K+ Apple Silicon + NVIDIA Blackwell**; optimising for ARM64 + CUDA.
- Real-time CV on **Jetson + DJI drone — 30fps** edge inference via RTMP.
- **LLM semantic translator** for **Schneider Electric** — IEC 61131-3 → IEC 61499 conversion automation for legacy PLC codebases; migration effort reduced by **70%**.
- **Technical document ingestion pipeline** (Schneider Electric) for machine / electronics engineering support — evolved from RAG + BM25 + keyword cluster indexing to trigram indices with custom pi.dev-based search agents. VLM + LLM stack across **2,000+ docs, 94% accuracy, 3× faster** than the prior solution.
- **Application security analysis platform** (Schneider Electric) — source + compiled-binary evaluation in computer-use sandboxes with OS state monitoring (registry / DLL deltas), pattern-matched against a custom-correlated **CVE / CWE / MITRE** index. Pentest lead time reduced from **3 weeks → 3 days.**
- **3 production AI systems; 2 peer-reviewed publications** in progress; team learnings published at blog.ecitis.org.

DevOps Engineer · *Pixa AI, Bengaluru* — Dec 2025

- Cloud infra + CI/CD for AI apps; **99.5% deployment success** across 2 Model API releases.

Web and Design Engineer · *Rossmann New Media 2 Inc.* — Jan 2026 – Present

- Worked on the website optimisation and redesign for rossmanngroup.com. Moved it into top 5 for SEO from being off the top 100.

Founding President · *PRISM* — Jan 2026 – Present

- Research club (AI/ML, SWE, HPC, DevOps, Biotech, InfoSec) — **150+ members.** Academic Advisory Board for AI Curriculum and Adoption.

Management Committee · *MIST* — Oct 2023 – Jun 2025

- CTF infra for **300+ users, 100% uptime**; cut costs by **80%.**
- **12+ workshops** for **150+ participants**; top 10% national CTFs.

RESEARCH

FL + Digital Twin for Route Optimisation · *SCE, MAHE* — Jun 2024 – Jul 2025

- GPS telemetry (**500K+ datapoints**); dashcam CV via OpenCV + YOLO — **12,000+ labelled samples.**

PROJECTS

Homelab Infrastructure · *Self-hosted cloud* — 2022 – Present

- **70+ services** on Proxmox/Docker — **98% uptime, 4 yrs**, 8TB MergerFS, Blackwell-GPU, 3-2-1 backups.
- Fine-tuned models for security (**30fps**) + 4K transcoding; **95%** CPU to GPU offload.
- Mitigated **live DDoS (50K+ req/min)**; **95%** attack surface reduction.

The TIK Menu API · GitHub — 2024 – 2025

- **5,000+ DAU, 150K+ monthly req**, campus adoption in under 7 days. REST API + Cloudflare R2.

Coolstuff.work Forgejo — 2023 – Present

- Suite of utility tools solving everyday problems for peers — **10,000+ aggregate** DAU across sub-apps.
- OR-Free: Free AI model tracker on OpenRouter; real-time availability and pricing dashboard.
- Contact Cards + Mass Contact Maker: vCard QR generator + bulk VCF creation from Excel sheets.

- Campus Directory: Crowdsourced campus contact hub with real-time status and fuzzy search.
- MIT Courses Search: Course discovery with interactive physics graph (**500+ nodes**), fuzzy search. Next.js 15, Radix UI, Fuse.js.
- transcript-cli: Multi-runtime video transcription CLI tool.
- **Sindhi Menu**: Cyclic mess menu viewer.

ophoner · *Mobile agent control loop* 2026 – Present

- Phone-based agent loop for device APIs and filesystem workflows — extensible via **tmux** and Android shell tooling. Focused on on-the-go file editing/querying without heavy desktop dependencies.

Developer Tooling (OSS) 2025 – Present

- agentskills: Published library of Claude Code skills for reusable agentic development workflows.
- t3code (fork): customised for parallelised agent-driven development flows.
- Self-hosted CI + sandboxing stack (migrating from Daytona); CodeRabbit integrated for public-repo review.